Final Lab Task Submission

Artificial Intelligence — Final Lab Task

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Section:	C	**Semester:**	Summer 25-26

Task 1 — Goal-Based Agent: Automated Water Purification System

The agent reads the water sensors (Turbidity, TDS, Bacteria, pH) and keeps applying **condition–action rules** — Sedimentation, Filtration, Reverse Osmosis, UV Treatment, Chemical Addition — printing the updated state after every action, until the goal state (Turbidity ≤ 1, TDS ≤ 300, Bacteria = False, pH 6.5–8.5) is reached.

Code Screenshot

task1_water_purification_agent.py - AI Lab - Visual Studio Code

task1_water_purification_agent.py ×

```python
# Final Lab Task 1
# Goal-Based Agent : Automated Water Purification System


def show(w):
    return (f"Turbidity = {w['Turbidity']}, TDS = {w['TDS']}, "
            f"Bacteria = {w['Bacteria']}, pH = {w['pH']}")


def goal_reached(w):
    if w["Turbidity"] <= 1 and w["TDS"] <= 300 and w["Bacteria"] == False:
        if 6.5 <= w["pH"] <= 8.5:
            return True
    return False


# sensor readings of the incoming water
water = {"Turbidity": 5.0, "TDS": 700, "Bacteria": True, "pH": 5.8}

print("Initial water quality:", show(water))

# keep applying the rules until the goal state is reached
while not goal_reached(water):

    if water["Turbidity"] > 3:
        print("→ Turbidity > 1 → Action Taken: Sedimentation")
        water["Turbidity"] = round(water["Turbidity"] * 0.1, 2)

    elif water["Turbidity"] > 1:
        print("→ Turbidity > 1 → Action Taken: Filtration")
        water["Turbidity"] = round(water["Turbidity"] * 0.4, 2)

    elif water["TDS"] > 300:
        print("→ TDS > 300 → Action Taken: Reverse Osmosis")
        water["TDS"] = int(water["TDS"] * 0.4)

```

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task1_water_purification_agent.py - AI Lab - Visual Studio Code

task1_water_purification_agent.py

```python
    elif water["Bacteria"] == True:
        print("→ Bacteria = True → Action Taken: UV Treatment")
        water["Bacteria"] = False

    elif water["pH"] < 6.5:
        print("→ pH < 6.5 → Action Taken: Add Chemicals")
        water["pH"] = 7.2

    elif water["pH"] > 8.5:
        print("→ pH > 8.5 → Action Taken: Add Chemicals")
        water["pH"] = 7.2

    print("   Updated state:", show(water))

print("Water quality after treatment:", show(water))
print("Goal Achieved: Water is purified and safe for drinking.")
```

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Output Screenshot

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\Minhaz\Desktop\AI Lab> python task1_water_purification_agent.py
Initial water quality: Turbidity = 5.0, TDS = 700, Bacteria = True, pH = 5.8
→ Turbidity > 1 → Action Taken: Sedimentation
   Updated state: Turbidity = 0.5, TDS = 700, Bacteria = True, pH = 5.8
→ TDS > 300 → Action Taken: Reverse Osmosis
   Updated state: Turbidity = 0.5, TDS = 280, Bacteria = True, pH = 5.8
→ Bacteria = True → Action Taken: UV Treatment
   Updated state: Turbidity = 0.5, TDS = 280, Bacteria = False, pH = 5.8
→ pH < 6.5 → Action Taken: Add Chemicals
   Updated state: Turbidity = 0.5, TDS = 280, Bacteria = False, pH = 7.2
Water quality after treatment: Turbidity = 0.5, TDS = 280, Bacteria = False, pH = 7.2
Goal Achieved: Water is purified and safe for drinking.
```

Task 2 — Simple Rule-Based Vacuum Cleaning Agent

In a 4×4 grid world (**D** = dirty, **.** = clean, **A** = agent) the agent **perceives** whether its current cell is dirty, **decides** its action (clean, move right, or move down to the next row) and **executes** it, printing the environment after every action until every cell is clean.

Code

```
# Final Lab Task 2
# Simple Rule-Based Vacuum Cleaning Agent

rows = 4
cols = 4

# 'D' = dirty cell , '.' = clean cell
grid = [
    ['D', '.', 'D', '.'],
    ['D', '.', 'D', '.'],
    ['D', '.', '.', '.'],
    ['D', 'D', 'D', 'D']
]


def show_grid(r, c):
    for i in range(rows):
        for j in range(cols):
            if i == r and j == c:
                print('A', end=' ')          # A = agent position
            else:
                print(grid[i][j], end=' ')
        print()
    print('--------------------')


row = 0
col = 0
count = 0

print("Initial Environment:")
show_grid(row, col)

while True:

    # rule 1 : if the current cell is dirty then clean it
    if grid[row][col] == 'D':
        grid[row][col] = '.'
        count = count + 1
        print(f"Cleaned cell at ({row}, {col})")
        show_grid(row, col)

    # rule 2 : otherwise move to the next cell
    if col < cols - 1:
        col = col + 1                        # move right
    elif row < rows - 1:
        row = row + 1                        # move down
        col = 0
    else:
        break                                # whole grid is scanned

    show_grid(row, col)

show_grid(row, col)
print("Cleaning complete. Total cells cleaned:", count)
```

Code Screenshot

task2_vacuum_cleaning_agent.py - AI Lab - Visual Studio Code

task2_vacuum_cleaning_agent.py

```python
# Final Lab Task 2
# Simple Rule-Based Vacuum Cleaning Agent

rows = 4
cols = 4

# 'D' = dirty cell , '.' = clean cell
grid = [
    ['D', '.', 'D', '.'],
    ['D', '.', 'D', '.'],
    ['D', '.', '.', '.'],
    ['D', 'D', 'D', 'D']
]


def show_grid(r, c):
    for i in range(rows):
        for j in range(cols):
            if i == r and j == c:
                print('A', end=' ')          # A = agent position
            else:
                print(grid[i][j], end=' ')
        print()
    print('--------------------')


row = 0
col = 0
count = 0

print("Initial Environment:")
show_grid(row, col)

```

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task2_vacuum_cleaning_agent.py - AI Lab - Visual Studio Code

task2_vacuum_cleaning_agent.py ×

```python
while True:

    # rule 1 : if the current cell is dirty then clean it
    if grid[row][col] == 'D':
        grid[row][col] = '.'
        count = count + 1
        print(f"Cleaned cell at ({row}, {col})")
        show_grid(row, col)

    # rule 2 : otherwise move to the next cell
    if col < cols - 1:
        col = col + 1                        # move right
    elif row < rows - 1:
        row = row + 1                        # move down
        col = 0
    else:
        break                                # whole grid is scanned

    show_grid(row, col)

show_grid(row, col)
print("Cleaning complete. Total cells cleaned:", count)
```

⎇ 0 ⚠ 0   Ln 55, Col 1   Spaces: 4   UTF-8   LF   Python 3.11.4 64-bit

Output Screenshot

Console output, captured in three parts (scrolled).

PROBLEMS   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS

```
PS C:\Users\Minhaz\Desktop\AI Lab> python task2_vacuum_cleaning_agent.py
Initial Environment:
A . D .
D . D .
D . . .
D D D D
--------------------
Cleaned cell at (0, 0)
A . D .
D . D .
D . . .
D D D D
--------------------
. A D .
D . D .
D . . .
D D D D
--------------------
. . A .
D . D .
D . . .
D D D D
--------------------
Cleaned cell at (0, 2)
. . A .
D . D .
D . . .
D D D D
--------------------
. . . A
D . D .
D . . .
D D D D
--------------------
. . . .
A . D .
D . . .
D D D D
--------------------
Cleaned cell at (1, 0)
. . . .
A . D .
D . . .
D D D D
--------------------
. . . .
. A D .
D . . .
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
D D D D
--------------------
. . . .
. . A .
D . . .
D D D D
--------------------
Cleaned cell at (1, 2)
. . . .
. . A .
D . . .
D D D D
--------------------
. . . .
. . . A
D . . .
D D D D
--------------------
. . . .
. . . .
A . . .
D D D D
--------------------
Cleaned cell at (2, 0)
. . . .
. . . .
A . . .
D D D D
--------------------
. . . .
. . . .
. A . .
D D D D
--------------------
. . . .
. . . .
. . A .
D D D D
--------------------
. . . .
. . . .
. . . A
D D D D
--------------------
. . . .
. . . .
. . . .
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
A D D D
--------------------
Cleaned cell at (3, 0)
. . . .
. . . .
. . . .
A D D D
--------------------
. . . .
. . . .
. . . .
. A D D
--------------------
Cleaned cell at (3, 1)
. . . .
. . . .
. . . .
. A D D
--------------------
. . . .
. . . .
. . . .
. . A D
--------------------
Cleaned cell at (3, 2)
. . . .
. . . .
. . . .
. . A D
--------------------
. . . .
. . . .
. . . .
. . . A
--------------------
Cleaned cell at (3, 3)
. . . .
. . . .
. . . .
. . . A
--------------------
. . . .
. . . .
. . . .
. . . A
--------------------
Cleaning complete. Total cells cleaned: 9
```